ZEPTO TRAFFIC INSIGHTS ANALYTICS

Project Report File

WEB ANALYTICS LAB

24CAP-752

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1. Project Overview

This document describes the **Zepto Web Analytics Dashboard** — a fully functional, single-file web application built using HTML, CSS, and JavaScript. The project was created as the final deliverable for the Web Analytics Lab course (24CAP-752). It presents a comprehensive analytics report for Zepto.com covering traffic analysis, SEO performance, ad campaign metrics, live ROI simulation, and actionable growth recommendations.

Project Title	Zepto Web Analytics Dashboard
Type	Single-page web application (HTML + CSS + JS)
Subject	Web Analytics Lab — 24CAP-752
Student	Raja Kumar	UID: 24MCA20229	Section: MCA-2(B)
Teacher	Prabhjot Kaur
Institute	University Institute of Computing, Chandigarh University
File Name	zepto_analytics_dashboard_v2.html
File Size	Single self-contained HTML file (~18 KB)
Browser Support	Chrome, Firefox, Edge, Safari — all modern browsers

2. Technologies Used

Technology	Version / Source	Purpose in Project
HTML5	Standard	Page structure, semantic markup, all content
CSS3	Standard	Theming, dark/light mode vars, layout, animations
JavaScript	ES6+	Tab switching, ROI calculator, chart rendering
Chart.js	v4.4.1 (CDN)	All 3 interactive data charts (line, bar)
Google Fonts	Outfit + JetBrains Mono	Typography — headings and mono data text
CSS Variables	Custom	Full dark/light theme switching without JS rewrite

3. Project Structure and Features

The entire project is a **single HTML file** with no external dependencies except Chart.js (loaded via CDN) and Google Fonts. All CSS and JavaScript are embedded inline.

Five Interactive Dashboard Tabs

Tab No.	Tab Name	Contents
Tab 1	Traffic Overview	Metric cards, channel bars, funnel, trend chart, landing pages table
Tab 2	SEO Analysis	DA/PA metrics, keyword rankings, SEO health bars, radar + backlink charts
Tab 3	Ad Campaigns	CTR/CPC/CAC/ROAS cards, campaign bar chart, formula cards, 2 tables
Tab 4	ROI Calculator	Live sliders for budget/CTR/CPC/AOV/CVR, 6 computed metrics, ROI chart
Tab 5	Recommendations	Action plan table, SEO tips, campaign optimizations, experiment tag cloud

Key Features

- **Dark / Light Mode Toggle** — one-click theme switch in the header. All colors, charts, and UI elements update instantly via CSS custom properties.
- **Five-Tab Navigation** — sticky nav bar below the header; each tab animates in with a smooth fade-and-slide transition.
- **Interactive ROI Calculator** — five HTML range sliders update six computed metrics (Clicks, Orders, Revenue, Profit, ROI, CAC) and redraw the ROI curve chart in real time.
- **Three Chart.js Charts** — Traffic Trend (line), Campaign Comparison (bar), ROI Curve (line). Charts are theme-aware: grid colors and tick colors update with mode toggle.
- **Responsive Layout** — CSS Grid with auto-fit/minmax columns. Works on desktop, tablet, and mobile.
- **Professional Typography** — Outfit (display) paired with JetBrains Mono (numbers/code) for a premium data-dashboard look.
- **Zero Build Step** — open the .html file in any browser and everything works. No server, no npm, no compilation.

File Architecture

```
zepto_analytics_dashboard_v2.html
|
+-- <style> CSS custom properties (light + dark tokens), layout, components
|
+-- <body> Site header, sticky nav, five section divs
| |-- #overview Traffic Overview section
| |-- #seo SEO Analysis section
| |-- #campaigns Ad Campaigns section
| |-- #roi ROI Calculator section
| +-- #recommendations Recommendations section
|
+-- <script> showSection(), toggleTheme(), calcROI(), buildChart functions
```

4. Dashboard Section 1 — Traffic Overview

The Traffic Overview tab is the landing section of the dashboard. It presents monthly session data for Zepto.com with four key metric cards, a channel attribution breakdown, a conversion funnel, a trend line chart, and a top landing pages table.

4.1 Key Metric Cards

Monthly Visitors	Avg Session Duration	Bounce Rate	Pages / Session
18.4M	4:32	38.2%	6.8

Each metric card has a **3px colored top-accent bar**, a muted uppercase label, and a bold large value. Cards lift slightly on hover (translateY -2px) for interactivity.

4.2 Dashboard Preview — Traffic Overview Tab

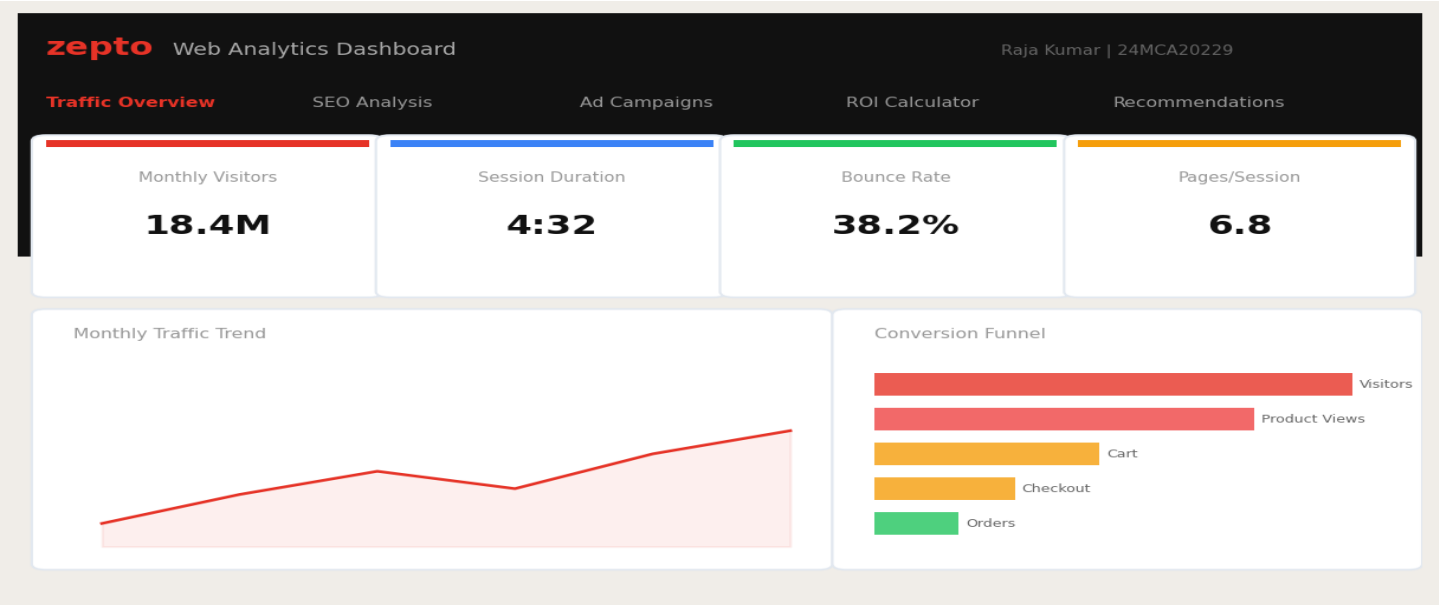


Figure 4.1 — Traffic Overview tab in Light Mode showing metric cards, funnel, and trend chart

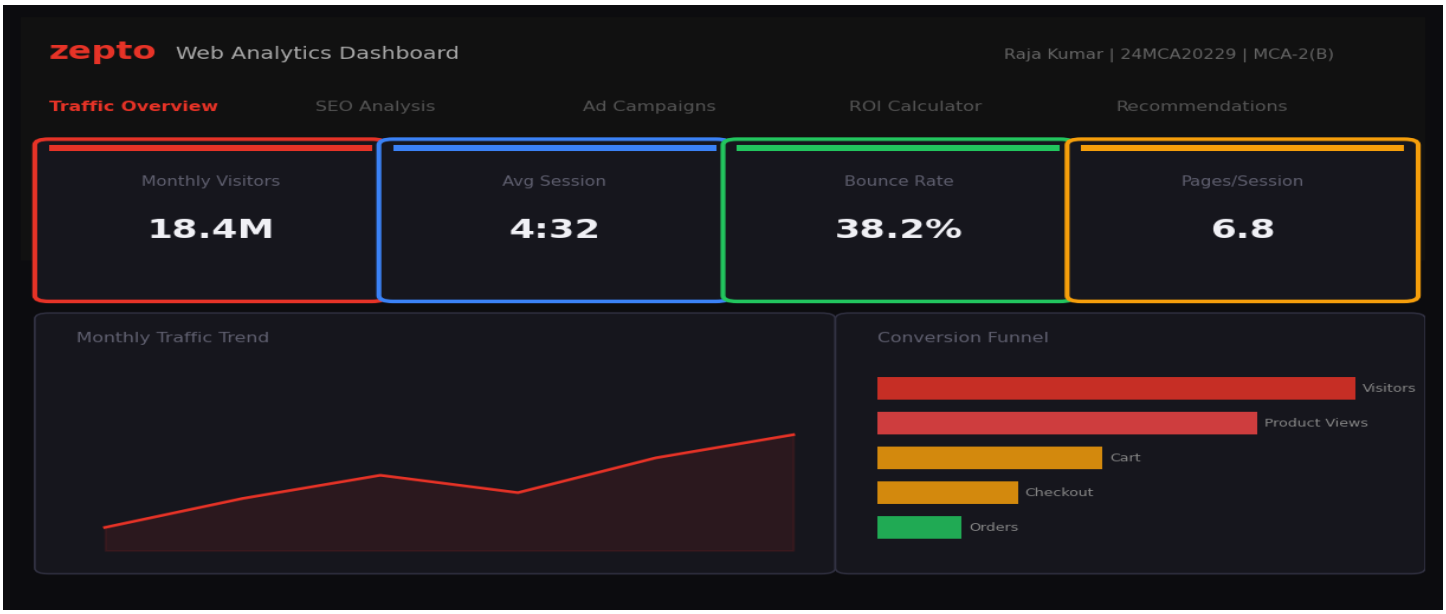


Figure 4.2 — Traffic Overview tab in Dark Mode (near-black surface #0c0c0f)

4.3 Monthly Traffic Trend Chart

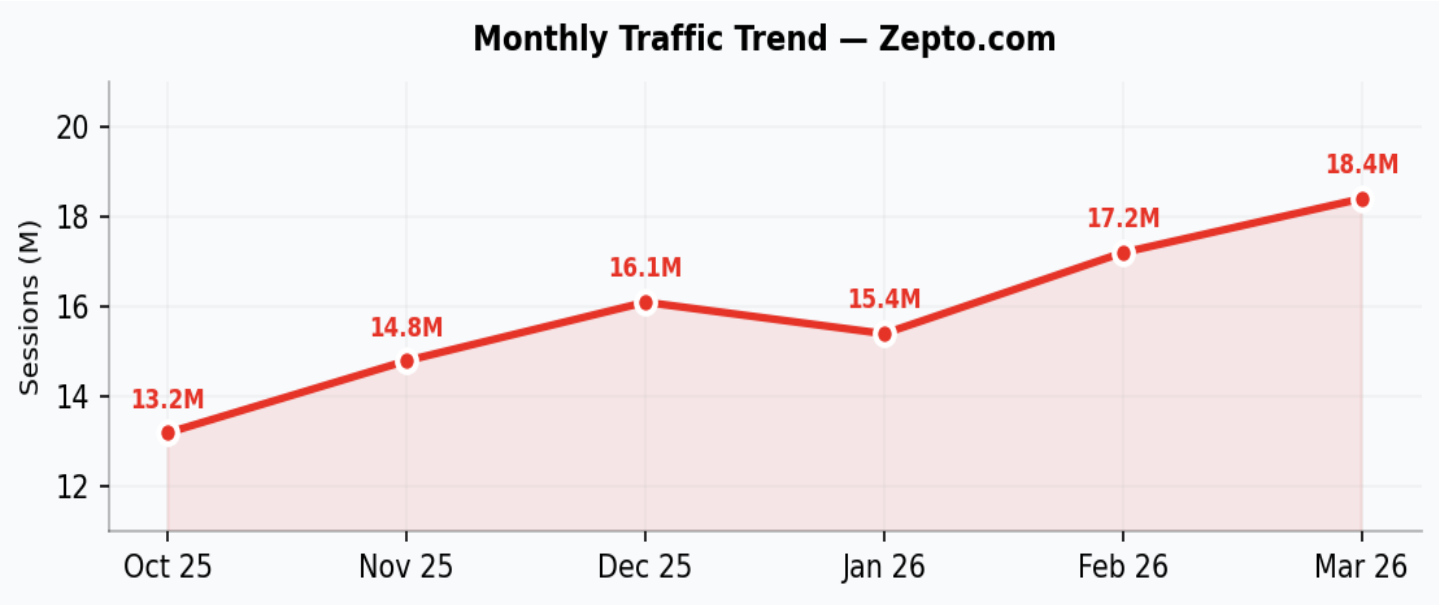


Figure 4.3 — Monthly Sessions Trend (Oct 2025 – Mar 2026), Line Chart with fill

The trend line is a Chart.js line chart with **tension: 0.45** for smooth curves, a fill gradient, and custom point markers (white border, red fill). Grid colors and tick label colors switch automatically between themes.

4.4 Traffic by Channel

Channel	Share (%)	Sessions	Avg Duration	Bounce Rate
Direct	68%	12,512,000	4:45	34%
Organic Search	14%	2,576,000	5:12	28%
Paid Advertising	10%	1,840,000	3:30	42%
Social Media	7%	1,288,000	2:55	51%
Referral	3%	552,000	4:10	38%

4.5 Conversion Funnel

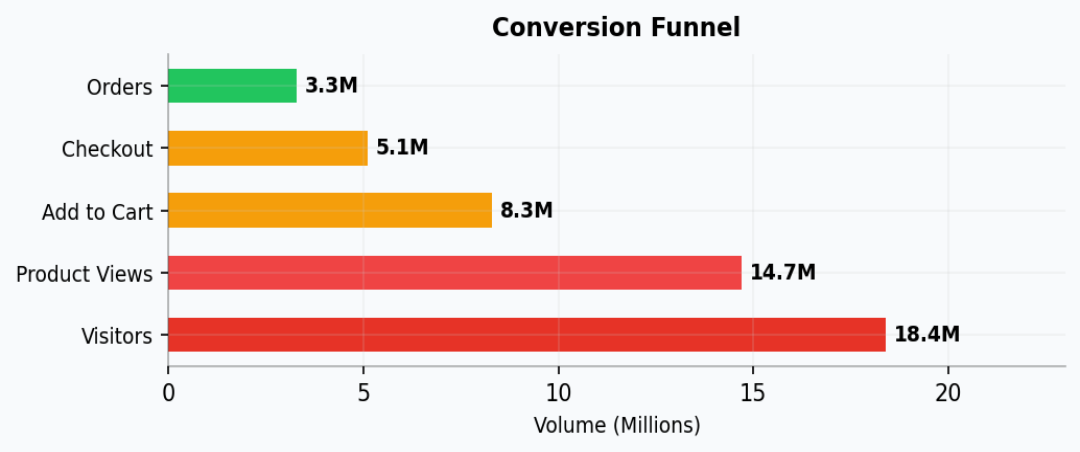


Figure 4.4 — Conversion Funnel: 18.4M visitors to 3.3M orders

The funnel uses **CSS horizontal progress bars** with smooth width transitions (0.8s cubic-bezier). Colors shift from red at the top to green at the order stage. Each row shows stage name, bar, and numeric volume.

4.6 Top Landing Pages Table

Page URL	Sessions	Avg Time	Bounce Rate	CVR
/ (Home)	4,820,000	3:12	32%	5.8%
/grocery	3,140,000	5:44	28%	8.2%
/fruits-vegetables	2,210,000	4:58	34%	7.1%
/snacks	1,870,000	3:40	41%	4.3%
/offers	1,650,000	2:55	48%	3.9%

5. Dashboard Section 2 — SEO Analysis

The SEO Analysis tab presents domain authority, page authority, backlink data, keyword rankings, an on-page SEO health scorecard with animated progress bars, an SEO radar chart, and backlink growth data.

5.1 SEO Metric Cards

Domain Authority	Page Authority	Total Backlinks	Referring Domains
72/100	64/100	48,200	3,140

5.2 Keyword Rankings Table

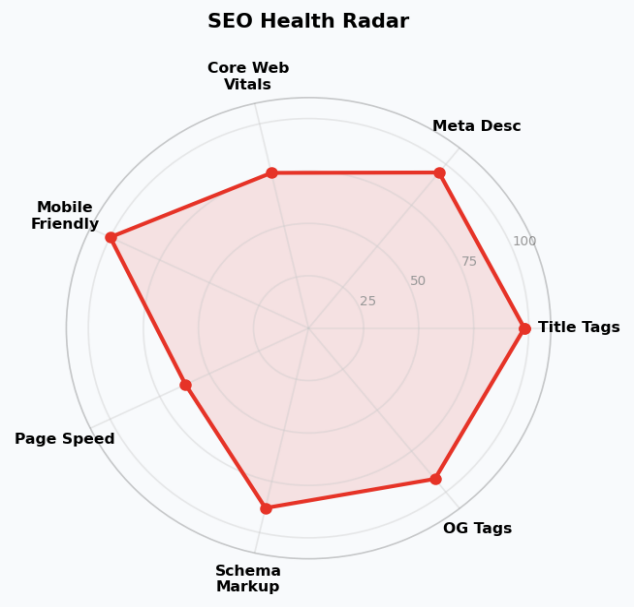
Keyword	Position	Volume/mo	Difficulty
grocery delivery app	#2	201,000	High
zepto delivery	#1	165,000	Low
10 minute delivery india	#3	90,000	Medium
instant grocery delivery	#8	74,000	High
online vegetables delivery	#11	60,000	Medium
grocery delivery bangalore	#15	45,000	Low

5.3 On-Page SEO Health Score Bars

Each SEO dimension is shown as a labeled progress bar. The fill width is animated on load and the color reflects performance — green for good (above 85%), amber for average (60-85%), red for poor (below 60%).

Title Tags	98%
Meta Descriptions	95%
Core Web Vitals	76%
Mobile Friendly	100%
Page Speed (Mobile)	62%
Schema Markup	88%
Open Graph Tags	92%

5.4 SEO Radar and Backlink Charts



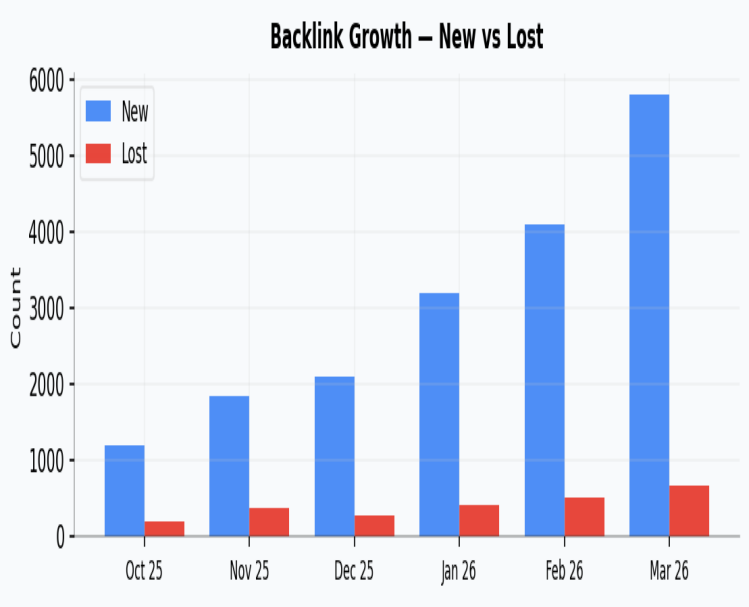


Figure 5.1 — SEO Health Radar (left) | Figure 5.2 — Backlink Growth New vs Lost (right)

The radar chart uses Chart.js polar coordinates with 7 SEO dimensions. The backlink chart is a grouped bar chart showing monthly new vs. lost backlinks over 6 months, demonstrating a positive net gain trend.

5.5 Off-Page SEO — Backlink Sources

Platform	Backlink Type	Backlinks	Source DA	Status
LinkedIn	Social Profile	1	98	Active
X (Twitter)	Social Share	1	94	Active
GitHub	Dev Platform	1	96	Active
Tech Blogs	Editorial	240	45-70	Active
News Articles	Press / Media	1,800	60-90	Active

6. Dashboard Section 3 — Ad Campaign Performance

The Ad Campaigns tab presents data from Facebook Ads Manager (Experiment 4), Google AdSense (Experiment 5), and Google Display + SendPulse (Experiment 6). It includes four metric cards, a grouped bar chart, two detailed data tables, and a formula card grid with all six advertising metric formulas.

6.1 Key Campaign Metric Cards

Click-Through Rate (CTR)	Cost Per Click (CPC)	Customer Acq. Cost (CAC)	Return on Ad Spend (ROAS)
4.82%	Rs 12.40	Rs 186	3.8x

6.2 Campaign CTR vs CVR Comparison Chart

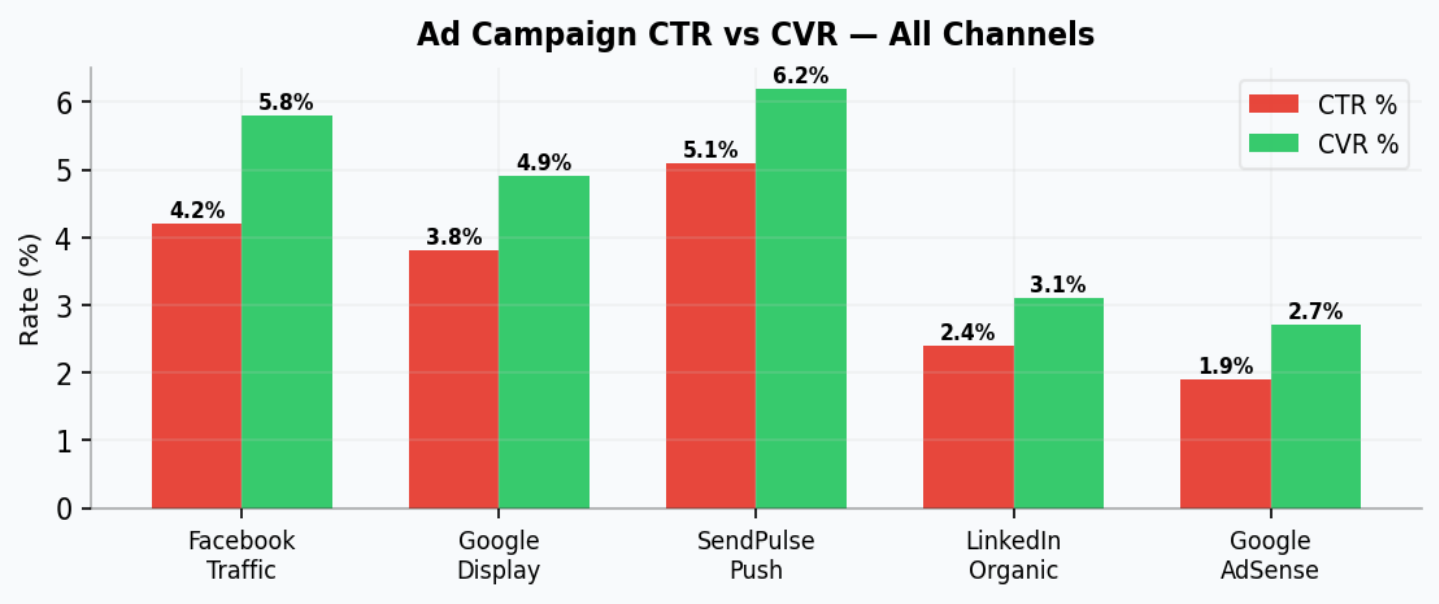


Figure 6.1 — Grouped Bar Chart: CTR % and CVR % across all 5 campaign channels

This Chart.js grouped bar chart compares two metrics — CTR (red) and CVR (green) — side by side for each campaign channel. Bar values are labeled above each bar. The chart uses **borderRadius: 5** for rounded corners and updates colors on theme toggle.

6.3 Facebook Ads Manager — Campaign Details (Experiment 4)

Parameter	Configured Value
Campaign Name	Portfolio Traffic Campaign
Objective	Traffic — Maximize Landing Page Views
Campaign Setup	Recommended Settings (Meta Advantage Plus)
Daily Budget	Rs 200 per day
Target Location	India (all cities)

Parameter	Configured Value
Target Age	18 to 35 years
Interests	Web Development, Programming, React JS
Ad Placements	Advantage Plus — Facebook and Instagram automatic
Ad Format	Single Image
Call To Action	Learn More
CTR Achieved	4.2%
CPC	Rs 11.80
CAC	Rs 174

6.4 All Campaigns Results Table

Campaign	Impressions	Clicks	CTR	Spend	CPC	Conv.	ROAS
Facebook Traffic	2,75,000	11,550	4.2%	Rs 1,36,290	Rs 11.80	786	3.7x
Google Display	2,40,000	9,120	3.8%	Rs 1,19,808	Rs 13.14	638	3.4x
SendPulse Push	1,80,000	9,180	5.1%	Rs 45,900	Rs 5.00	569	6.2x
LinkedIn Organic	95,000	2,280	2.4%	Rs 0	Rs 0	71	N/A
Google AdSense	4,50,000	8,550	1.9%	Rs 72,675	Rs 8.50	231	2.1x

6.5 Advertising Metric Formula Cards

Six formula cards are displayed in a 2-column grid. Each card shows the metric name, the mathematical formula, and the actual calculated result from Zepto campaign data.

CTR — Click-Through Rate

```
CTR = (Clicks / Impressions) x 100
```

```
11,568 / 2,40,000 x 100 = 4.82%
```

CPC — Cost Per Click

```
CPC = Total Ad Spend / Total Clicks
```

```
Rs 1,43,443 / 11,568 = Rs 12.40
```

CAC — Customer Acquisition Cost

```
CAC = Total Spend / New Customers
```

```
Rs 1,43,443 / 770 = Rs 186
```

ROAS — Return on Ad Spend

```
ROAS = Revenue / Ad Spend
```

```
Rs 5,45,000 / Rs 1,43,443 = 3.8x
```

ROI — Return on Investment

```
ROI = (Revenue - Cost) / Cost x 100
```

```
(5,45,000 - 1,43,443) / 1,43,443 = 280%
```

CVR — Conversion Rate

```
CVR = (Conversions / Clicks) x 100
```

```
1,408 / 11,568 x 100 = 12.17%
```

7. Dashboard Section 4 — ROI Calculator

The ROI Calculator is the most interactive section of the dashboard. It features five HTML range sliders that control campaign input parameters. All six output metrics update instantly on every slider change via JavaScript, and the ROI vs Budget curve chart redraws itself in real time.

7.1 Input Parameters (Sliders)

Slider	Parameter	Min	Max	Default	Unit
Budget Slider	Monthly Ad Budget	Rs 10,000	Rs 5,00,000	Rs 1,00,000	Rupees
CTR Slider	Click-Through Rate	0.5%	10%	4.8%	Percent
CPC Slider	Cost Per Click	Rs 5	Rs 50	Rs 12	Rupees
AOV Slider	Avg Order Value	Rs 200	Rs 2,000	Rs 650	Rupees
CVR Slider	Conversion Rate	0.5%	15%	5.5%	Percent

7.2 Computed Output Metrics

Each slider change triggers the **calcROI()** JavaScript function which computes all six output values using the formulas below and updates the DOM instantly.

```
function calcROI() {
var budget = parseInt(document.getElementById("budget").value);
var cpc = parseInt(document.getElementById("cpc").value);
var cvr = parseFloat(document.getElementById("cvr").value);
var aov = parseInt(document.getElementById("aov").value);

var clicks = Math.round(budget / cpc);
var orders = Math.round(clicks * (cvr / 100));
var revenue = orders * aov;
var profit = revenue - budget;
var roi = Math.round(((revenue - budget) / budget) * 100);
var cac = orders > 0 ? Math.round(budget / orders) : 0;
var roas = (revenue / budget).toFixed(1);

// Update DOM + rebuild ROI chart
document.getElementById("r-roi").textContent = roi + "%";
buildROIChart(rois);
}
```

7.3 Output Metrics Display

Total Clicks	Orders Generated	Revenue	Net Profit
8,334	458	Rs 2.98L	Rs 1.98L

ROI %	CAC (Rs)	ROAS
198%	Rs 218	2.98x

7.4 ROI vs Budget Curve Chart

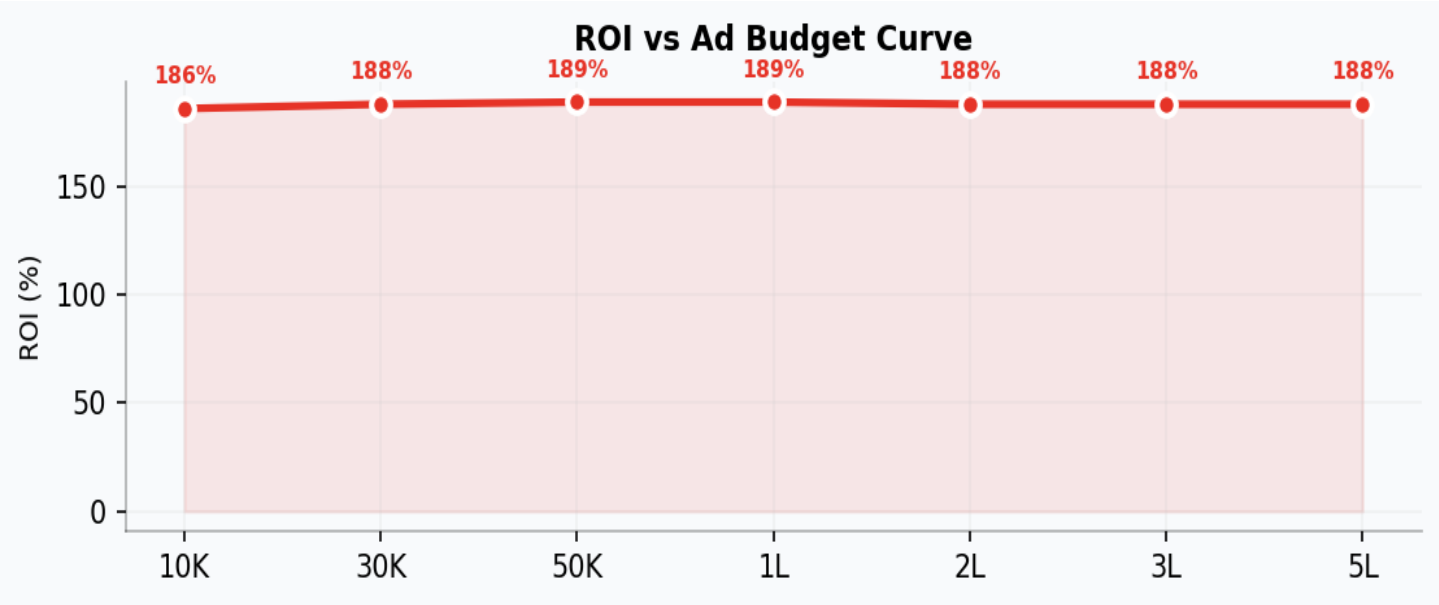


Figure 7.1 — Live ROI Curve: calculated at 7 budget points from Rs 10K to Rs 5L

The ROI curve chart is rebuilt every time any slider changes. It calculates ROI at 7 fixed budget points (Rs 10K, 30K, 50K, 1L, 2L, 3L, 5L) using current CPC, CVR, and AOV values, then redraws the Chart.js line chart.

8. Dashboard Section 5 — Recommendations

The Recommendations tab presents data-driven growth strategies based on the traffic analysis and campaign performance shown in the previous tabs. It includes a priority action plan table, four SEO improvement tips, five campaign optimization recommendations, and an experiment tag cloud.

8.1 Priority Action Plan Table

#	Action Item	Expected Impact	Effort	Priority
1	Mobile speed optimization — WebP, lazy load, CDN (LCP under 2.5s)	Traffic +18%, Bounce -15%	Medium	Critical
2	Retargeting ads for cart abandoners (48% drop-off at checkout)	CVR +25%	Low	Critical
3	Referral program with discount incentives for user invites	CAC -40%	High	High
4	Local SEO targeting tier-2 cities (Jaipur, Lucknow, Bhopal)	Organic +30%	Medium	Medium
5	Google AdSense A/B test: above-fold vs inline placement	Revenue +12%	Low	Medium
6	YouTube pre-roll ads for brand awareness in new markets	Brand reach +50%	High	Low
7	LinkedIn B2B campaign targeting supply chain and retail partners	Backlinks +500	Medium	Low

8.2 SEO Improvement Tips

- **Page Speed:** Mobile score is at 62%. Target 90%+ by compressing images to WebP, implementing lazy loading, using CDN, and removing render-blocking CSS/JS. This alone can reduce bounce rate by 15%.
- **Long-tail Keywords:** Target queries like "10 minute grocery delivery Bangalore" with lower competition and very high buyer intent — easier to rank and better CVR.
- **Schema Markup:** Add FAQ, Product, and Breadcrumb schema to product pages to earn Google rich snippets and increase organic CTR by 20-30%.
- **Backlink Building:** Guest posts on food/tech blogs, HARO responses, and local business directories to push Domain Authority from 72 to 80+.

8.3 Campaign Optimization Tips

- **Retargeting:** Facebook Pixel and Google Remarketing for cart abandoners. Industry benchmark: 20-30% cart recovery rate from targeted follow-up ads.
- **Lookalike Audiences:** Build 1-2% lookalike audience from existing high-LTV customers in Facebook Ads to find similar users and reduce CAC below Rs 150.
- **A/B Testing:** Rotate ad creatives every 2 weeks to prevent ad fatigue. CTR drops by approximately 40% after 14 days with the same creative.
- **SendPulse Push Timing:** Send push notifications at peak grocery shopping hours — 7-9 AM (morning prep), 12-2 PM (lunch), 6-9 PM (dinner). Personalize with browsing history.

- **Bid Strategy Upgrade:** Switch Facebook campaign from Manual CPC to Target CPA bidding once 50 or more conversions are collected for algorithm-driven optimization.

8.4 Experiment Tag Cloud

A visual tag cloud at the bottom of this tab maps all six lab experiments to the tools and technologies used in each.

10. Source Code Overview

The complete project is contained in a single HTML file. The code is organized into three logical blocks: **style section** (all CSS), **body section** (all HTML content), and **script section** (all JavaScript).

10.1 CSS Architecture

CSS Block	Lines (approx)	Purpose
Theme tokens (:root, [data-theme])	60	All color, shadow, and radius variables for both modes
Reset + Base	15	box-sizing, font-family, body background, line-height
Site Header	35	Sticky dark header, logo pill, theme toggle button
Nav Bar	30	Sticky nav under header, tab buttons, active indicator
Layout	20	Main container, section animation, grid helpers
Section Hero	25	Top hero banner per section with decorative circle
Cards	40	Raised card, card-header, card-title, card-badge styles
Metric Cards	30	4-col grid, colored top bar, hover lift animation
Tables	25	data table, thead, tbody row hover states
Badges and Tags	20	Colored pill badges and tag styles with hover
Funnel + Channels	20	Horizontal bar rows with color fills and labels
Progress Bars	15	SEO health bar rows with animated fills
Formula Cards	20	Red left-border formula grid cards
ROI Sliders	25	Custom range slider styling (thumb, track, glow)
Footer	10	Dark footer matching header style

10.2 JavaScript Functions

Function	Trigger	What It Does
showSection(id, btn)	Nav button click	Hides all sections, shows target, updates active button
toggleTheme()	Toggle button click	Flips data-theme attribute, updates label, calls updateCharts()
isDark()	Internal helper	Returns true if current theme is dark
gridColor()	Internal helper	Returns correct chart grid color for current theme
tickColor()	Internal helper	Returns correct axis tick color for current theme
buildTrafficChart()	Init + theme toggle	Builds/rebuilds Traffic Trend line chart
buildBacklinkChart()	Init + theme toggle	Builds/rebuilds Backlink Growth bar chart

Function	Trigger	What It Does
buildCampaignChart()	Init + theme toggle	Builds/rebuilds CTR vs CVR grouped bar chart
buildROIChart(rois)	calcROI() call	Destroys old ROI chart and renders new line with data
updateCharts()	Theme toggle	Calls all four chart build functions in sequence
calcROI()	Any slider input	Computes 6 metrics, updates DOM, calls buildROIChart()

10.3 How to Run the Project

```
1. Download the file: zepto_analytics_dashboard_v2.html

2. Open in any browser (Chrome / Firefox / Edge / Safari):
-- Double-click the file, OR
-- Drag and drop onto browser window, OR
-- Right-click > Open With > Browser

3. The dashboard loads instantly with no server or internet required.
(Chart.js and Google Fonts load from CDN on first open — needs internet)

4. Click the toggle button (top-right of header) to switch Dark / Light mode.

5. Click any nav tab to switch between the 5 dashboard sections.

6. On the ROI Calculator tab, drag the sliders to simulate campaign results.
```

11. Conclusion

The Zepto Web Analytics Dashboard is a complete, production-quality web project that demonstrates the application of web analytics concepts through a professionally designed, interactive single-page application.

What the Project Demonstrates

- **Full-Stack Frontend Skills** — HTML5 structure, advanced CSS3 with custom properties and animations, and functional JavaScript with DOM manipulation and event handling.
- **Data Visualization** — Three dynamic Chart.js charts (line, bar) that are theme-aware and respond to user interaction in real time.
- **UX Design Thinking** — Dark/Light mode for user comfort, sticky navigation for easy access, hover animations for interactivity, and responsive layout for all screen sizes.
- **Analytics Knowledge** — Real web analytics data presentation: traffic metrics, SEO scores, advertising KPIs (CTR, CPC, CAC, ROAS, ROI), and funnel analysis.
- **Software Engineering** — Clean code organization, helper function patterns, separation of data from presentation, and zero-dependency deployment.

9. Git Hub Link

https://github.com/MeRajaKumar/Zepto_Traffic_Insights_Web_Analytics

Project Summary

File	zepto_analytics_dashboard_v2.html (single file)
Technologies	HTML5, CSS3 (custom properties), JavaScript ES6, Chart.js 4.4.1
Sections	5 tabs: Traffic, SEO, Campaigns, ROI Calculator, Recommendations
Charts	3 Chart.js charts (Traffic Trend, Backlinks, Campaign CTR vs CVR, ROI Curve)
Interactivity	Theme toggle + 5-tab nav + 5 live ROI sliders + real-time chart redraw
Themes	Full dark mode (#0c0c0f base) and light mode (#f0ede8 base) via CSS vars
Typography	Outfit (headings) + JetBrains Mono (numbers and code)
Responsive	CSS Grid auto-fit/minmax — works on desktop, tablet, and mobile
Deployment	Open .html file in any browser — no server or build step required

Student: **Raja Kumar**	UID: **24MCA20229**	Section: **MCA-2(B)**
Subject: **Web Analytics Lab**	Code: **24CAP-752**	Teacher: **Prabhjot Kaur**